

Sophie Choi

Secret Clearance | 626-734-3575 | sophie.elissa.choi@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of California, Irvine

Irvine, CA

Bachelor of Science in Computer Science, GPA: 3.80

WORK EXPERIENCE

Software Developer Intern

Jun 2026 – Aug 2026

Naval Air Warfare Center Weapons Division (NAWCWD)

Point Mugu, CA

- Co-authored a bidirectional real-time **C#/.NET** application (**WPF, MVVM**) that transmits positioning commands to a \$6M Weibel radar system and parses live radar telemetry from 9 targets into human-readable format in real-time, streamlining military range testing operations
- Developed a **C#/.NET** system configuration parser to validate **network socket connections** at startup and authored **OOP** data structures that bridge frontend display and backend logic, ensuring the application launches reliably before real-time operations begin
- Presented system architecture, **UML** class diagrams, and **XAML GUI** designs at critical design reviews, aligning hardware stakeholders on implementation decisions before development milestones
- Authored the full unit test suite in **C#**, covering end-to-end **network connectivity** and wrote application test procedure documents to standardize **SDLC** validation

EXPERIENCE

Co-Infrastructure Lead

Oct 2025 – Present

Cyber @ UCI Infrastructure Team

Irvine, CA

- Maintained **SSO (Authentik)**, **IaC tooling**, **Proxmox** virtualization with **Linux**, and **Docker** containerization across a production network serving 100+ club members, ensuring up-time and security for cyber team trainings and workshops
- Reduced cluster overhead by 25% and improved operational efficiency by doing routine maintenance on the **Proxmox** hypervisor server cluster (**Linux**) and reorganizing **NFS storage** configurations
- Improved storage bandwidth to 40 GB/s by configuring a dedicated storage switch connecting **Proxmox** hypervisor nodes to a **NAS** using **CLI**-based network configuration, isolating storage traffic from the production network
- Leading architecture and **Figma** UI/UX design for 4 internal tools built with **React**, **GitLab**, and **Docker**, including a VM provisioning tool serving 20+ internal club members
- Deployed **Proxmox VE** high-availability clustering and load balancing, eliminating single points of failure across the cluster to ensure uninterrupted access to club infrastructure

Full-Stack Developer

Oct 2025 – Present

Commit the Change

Irvine, CA

- Programmed a full-stack application using **React**, **Express.js**, **PostgreSQL**, and **CSS**, enabling scalable data management and real-time program oversight for a global NPO operating in 5 continents
- Deployed **RESTful APIs in JavaScript** with **full CRUD functionality** and Promise-based concurrent data fetching, streamlining frontend-backend data flow and improving platform performance by ~25%
- Collaborated in **Agile** sprints with pair programming and **Git** to ship **React** features against 2-week deadlines

Software Developer

Apr 2026 – Present

Blockchain @ UCI Smart Contract Team

Irvine, CA

- Engineered an AI-powered exploit generation engine using **Python**, **RESTful APIs**, and **Anvil local chain**, delivering binary proof-of-concept attacks on live contract state rather than theoretical vulnerability reports
- Built a **Python Slither static analysis module** detecting reentrancy, access control, and logic flaws in a smart contract, structuring findings as JSON to feed a downstream **LangGraph agent pipeline**
- Containerized the platform with **Docker** and designed a shared **Postgres/pgvector** database architecture, giving all developers a consistent environment and eliminating data inconsistencies across the team

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, C#, HTML, CSS, XML

Frameworks/Libraries: React, Node.js, Express, PostgreSQL, .NET, WPF, MVVM, Tailwind CSS

Tools: Docker, Proxmox, IaC, Linux, NAS Administration, Git, GitLab, Figma, Firebase, Unity, Microsoft Office 365

Certifications: Cisco Certified Network Associate (CCNA), Girls Who Code: Data Science for Good